Unit (1): Lesson (1)

Divisibility

Complete:

1) **The largest 3-digit number divisible by 2, 5 and 10 is …….**
2) **If 5 X 4 = 20, then 20 is a ……. of 5 and 4.**
3) **If 6 X 8 = 48, then 6 and 8 are ……. of 48.**
4) **If 32 is a multiple of 8, then 32 is ……. by 8.**
5) **The smallest 3-digit number divisible by 2, 4 and 10 is …….**
6) **If 54 is divisible by 6, then 6 is called a ……. of 54.**
7) **For any number to be divisible by 6, it must be divisible by …… and ……**
8) **For any number to be divisible by 10, it must be divisible by …… and ……**
9) **8 is a ……. of 72.**
10) **For any number to be divisible by 10, its ones digit must be …….**
11) **35 is ……. by 7.**
12) **4 is a ……. of 2.**
13) **3 is a ……. of 27.**
14) **If 5 X 6 = 30, then 30 is a ……. of 6 and 5 is ……. of 30.**
15) **From the multiples of 8 ……. and …….**
16) **From the factors of 12 ……. and …….**
17) **24 is divisible by ……. and …….**

Answer the following:

1) **A school took 228 students in a trip. Can we distribute them equally among 4 buses? Why?**

_____
_____
_____
_____
_____

2) **The food bank wants to distribute 567 boxes. It is possible to distribute the boxes among 6 villages equally ?**

_____
_____
_____
_____
_____

3) **Sara has 105 toys, can she divide them among 10 of her friends? Why or why not ?**

_____
_____
_____
_____
_____

4) **Mom bought 18 pencils, and she wants to divide them among her 3 sons. Can she ? Why?**

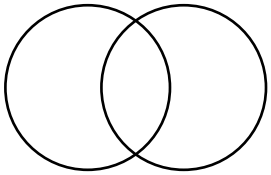
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Unit (1): Lesson (2)

Factorizing a Number to its Prime Factors

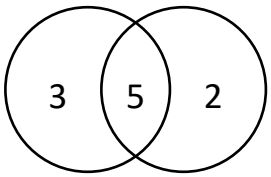
Answer the following:

1) Find the GCF and LCM of 15 and 45 using Venn Diagram:



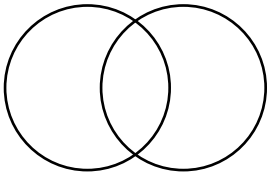
a. $15 =$
b. $45 =$
c. $GCF =$
d. $LCM =$

2) Find the GCF and LCM of 15 and 45 using Venn Diagram:



a. The two numbers represented in the venn diagram are ……. and ……
b. The GCF for the two numbers are ……
c. The LCM for the two numbers are ……

3) Find the GCF and LCM of 12 and 30 using Venn Diagram



a. $12 =$
b. $30 =$
c. $GCF =$
d. $LCM =$

Complete:

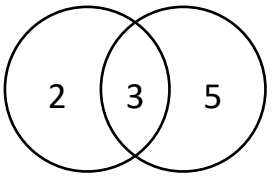
1) The common factor of all numbers is ……

a. 0 **b. 1** **c. 2** **d. 3**

2) The lowest common multiple of any two prime numbers is ……

a. 0 **b. 1** **c. 2** **d. 3**

3) From the opposite venn diagram, the LCM is ……



a. 5 **b. 3**

c. 30 **d. 2**

4) The greatest common factor of 4 and 20 is ……

a. 1 **b. 2** **c. 4** **d. 20**

5) The greatest common factor of 5 and 7 is ……

a. 1 **b. 2** **c. 3** **d. 6**

6) From the opposite venn diagram, the GCF is ……

a. 2 **b. 21** **c. 14** **d. 7**

7) The only even prime number is …….

a. 0 **b. 4** **c. 2** **d. 3**

8) The smallest odd prime number is ……..

a. 1 **b. 3** **c. 0** **d. 2**

9) …….. is neither prime nor a composite number.

a. 1 **b. 3** **c. 2** **d. 0**

10) The LCM of 3 and 6 is ……..

a. 6 **b. 3** **c. 1** **d. 18**

Unit (1): Lesson (3)

Writing Expressions Using the G.C.F

Complete:

1) 15 + 5 = 5 (------ +)

2) 3 × (------ +) = (15 + 9)

3) 21 + 28 = 7 × (------ +)

4) ------ × (------ +) = 2 × 8 + 2 × 9

5) 4(6 + 5) = ------ × ------ + ------ × ------

6) 40 + 60 = ------ (------ + ------)

7) 7 × (------ + ------) = (------ × 8) + (- × 1)

Use the distributive property and GCF to write the expression of the following addition Problems:

1) 25 + 35

_____

_____

_____

_____

_____

2) Mariam prepared bags containing snacks of 12 bananas and 18 cupcakes, she wants the snacks in the bags to be distributed evenly without any food left. What is the greatest number of bags of snacks can she prepare? How many bananas and cupcakes are there in each bag ?

_____

_____

_____

_____

_____

Unit (1): Lesson (4)

Analyzing Least Common Multiples

Find the result of the following in the simplest form:

1) $4\frac{1}{8} + 3\frac{3}{4} =$ ……

2) $9\frac{3}{4} - 2\frac{2}{5} =$ ……

3) $4\frac{2}{3} + 5\frac{1}{4} =$ ……

4) $10\frac{5}{6} - 4\frac{1}{9} =$ ……...

5) $6\frac{1}{12} + 4\frac{1}{18} =$ ……..

Unit (1): Assessment

Choose the correct answer from those given:

1) $384 \div 16 = 24$, the dividend is …….

a. 348 b. 16 c. 24 d. 0

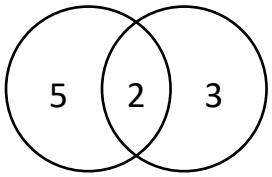
2) Sara distributed 120 cookies equally among 6 boxes, how many cookies are left?

a. 0 b. 1 c. 2 d. 3

3) The common factor of all numbers is ……..

a. 0 b. 1 c. 2 d. 3

4) The opposite Venn Diagram represents the prime factorization of 2 numbers which are …….



a. 3 and 5 b. 2 and 3

c. 2 and 5

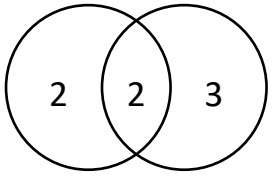
d. 6 and 10

5) The GCF of any two prime numbers is …….

a. 0 b. 2

c. 3 d. 1

6) The following represents the greatest number of bags that can be made from apples and bananas respectively, the number of all the bags is ……



a. 4 b. 2

c. 6 d. 3

7) ……. $\times (5 + 1) = (7 \times 5) + (7 \times 1)$

a. 6 b. 35 c. 1 d. 7

Answer the following:

1) The two numbers represented in the venn diagram are …… and …….

a. The common prime factors of the two numbers are ……..

b. The GCF of the two numbers is ……..

c. The GCF of the two numbers is ……...

2) Adam prepared bags containing snacks, he has 6 oranges and 18 pieces of fruit, if he wants the snacks to be distributed equally. How many bags can he prepare?

_____

_____

_____

_____

_____

3) Find the GCF and LCM of 12 and 18 by factorization

_____

_____

_____

_____

_____

4) Use Venn Diagram to find the GCF and LCM of:

a. 15 and 10

_____

_____

b. 27 and 4

_____

_____

5) <u>Complete:</u>

a. **If $6048 \div 24 = 252$, then 24×252 = ----------**

b. **The smallest prime number is ----------**

c. **The smallest odd prime number is ----------**

d. **The only even prime number is ----------**

e. **The prime factors of 15 are ----------**

f. **The greatest 2-digit prime number is ----------**

g. **The prime number has only ---------- factors**

h. **All prime numbers are odd except ----------**

i. **The common multiple of all numbers is ----------**

j. **The common factor of all numbers is ----------**

k. **The number whose prime factors are 2,3 and 7 is ----------**

l. **The prime factors of 24 are ----------**

m. **The GCF of 2 prime numbers is -------**

n. **$5 \times$ (----- + -----) = (----$\times 2$) + (-----$\times 4$)**

o. $2\frac{1}{3} + 1\frac{1}{4} =$ ----------

p. ------$+1\frac{2}{3} = 2\frac{1}{4}$

q. $3\frac{1}{2} -$ --------- $= 1\frac{3}{8}$

6) The students collected 20 boxes of cheese and 40 bags of legumes. What is the largest number of baskets of food that can be prepared without any leftover?

_____
_____
_____
_____
_____

7) Karim has 48 pencils and 18 crayons. What is the greatest number of sets that can be made to have the same number of each set?

_____
_____
_____
_____
_____

Unit (2): Lesson (1&2)

Using a Number Line to Describe Data

Using a Number Line and Symbols to Compare Numbers

1) Put <, > or =:

a) 8 ----- 7 **b) 4 ----- 5** **c) 2 ----- 1**

d) -6 ----- 4 **e) -2 ----- 3** **f) -45 ----- 9**

g) -8 ----- (-9) **h) –(-3) ----- (-8)** **i) 3 ----- -3**

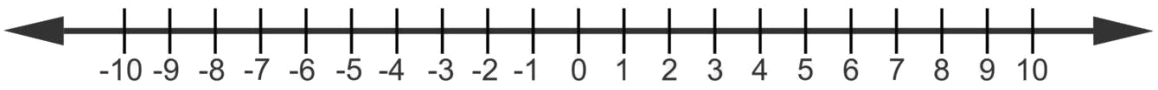
2) Arrange the following in ascending order:

6 , -7 , 3 , -5 , 1 , 8 , -9 and -2

…….., ……. ,……. ,……. ,……. , ……. ,……. ,…….

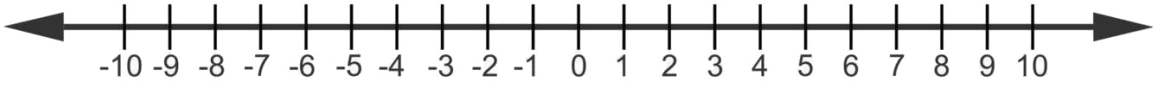
3) Represent the following on the number line:

5, -3, 1 , -5, 0, -9



4) Represent each number and its inverse on the number line:

a) 6 **b) -7** **c) 2**



5) Complete:

a. **The smallest positive number is --------- while the greatest negative number is --------**

b. **The smallest odd number is --------- while the smallest even number is -------**

c. **The smallest natural number is --------- while the smallest counting number is --------**

d. **The integers between -6 and 3 are -------**

e. **The integer just after 0 is --------**

f. **The integer just before -16 is ---------**

g. **The distance between -1 and 4 on the number line equals ----------**

6) Write the additive inverse of the following:

a. **9 -----**

b. **-6 -----**

c. **0 -----**

d. **0.6 -----**

e. **82 -----**

f. **-14 -----**

g. **-3 -----**

h. **34 -----**

i. **1 -----**

j. **9.3 -----**

k. $\frac{1}{8}$ **-----**

l. **−0.5 -----**

Unit (2): Lesson (3)

Analyzing Rational numbers by using models

1) Write the following Rational numbers in the form $\frac{a}{b}$

a. -9 ……. **d.** 45 …….

b. -0.39 ……. **e.** -3.17 …….

c. $5\frac{1}{3}$……. **f.** $2\frac{1}{4}$ …….

2) Match the numbers to the best subset:

(Rational number- Integer – Counting numbers – Natural numbers)

a. $0.5 =$ ……. **e.** $6 =$ ……. **i.** $100 =$ …….

b. $-2 =$ ……. **f.** $-3 =$ ……. **j.** $-3\frac{1}{2} =$ …….

c. $0 =$ ……. **g.** $18 =$ ……. **k.** $1 =$ …….

d. $\frac{4}{9} =$ ……. **h.** $-41 =$ ……. **l.** $-2.3 =$ …….

3) Choose:

1. The best subset for the number 8 is a/an …… number

a. Counting **b. Natural** **c. Integer** **d. Rational**

2. The smallest positive integer is …….

a. -99 **b. 0** **c. -1** **d. 1**

3. $-\left|-\frac{10}{2}\right|$ **belongs to the set of ……. Numbers**

a. Counting **b. Natural** **c. Integer** **d. Rational**

4. Which of the following is the smallest number? ……

a. -1.6 **b. -6.1** **c. -1** **d. 0**

5. The best subset of the number 7 is …… number

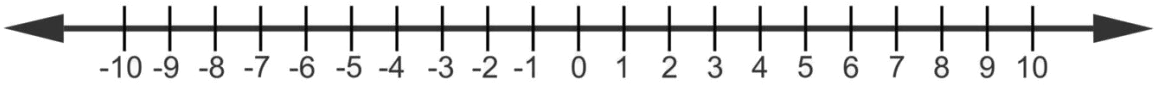
a. Counting **b. Natural** **c. Integer** **d. Rational**

3) Represent the following on the number line:

5, -3, 1 , -5, 0, -9

4) Represent the following on the number line:

a. $3\frac{1}{2}, 1.4, -6\frac{1}{3}, -5, 0, -8$



5) Write belongs or doesn't belong:

a. $\frac{1}{5}$ **…….. the set of integers**

b. 0 **…….. the set of rational numbers**

c. -9 **…….. the set of counting numbers**

d. $\frac{12}{6}$ **…….. the set of counting numbers**

e. $\frac{0}{4}$ **…….. the set of natural numbers**

Unit (2): Lesson (4)

Comparing and Ordering Rational Numbers

1) Complete:

a. The rational number between -5.6 and -5.4 is ……...

b. The number $-3\frac{1}{5}$ in the form $\frac{a}{b}$ is ……

c. The opposite of 0 is ……..

d. -3.6 < ……..

e. The smallest natural number is ……..

f. 0.3 > ……..

2) Arrange the following form the least to the greatest:

$0.15, 0.63, -0.4, -0.34, 0.1$

……. , …… , ……. , …… , ……. ,

3) Arrange the following form the least to the greatest:

$4.6, 4\frac{1}{5}, -3, -3\frac{1}{2}, -12$

……. , …… , ……. , …… , ……. ,

4) Write a rational number lying between the following:

a. $\frac{1}{5}$ …….. $\frac{3}{5}$

b. $\frac{1}{4}$ …….. $\frac{3}{8}$

c. (-7.2) …….. (-7)

d. $\frac{1}{4}$ …….. $\frac{1}{2}$

Unit (2): Lesson (5&6)

Exploring Absolute Value

Comparing Absolute Values

1) Find the value of the following:

a. $|-2| = \ldots\ldots$

b. $-|11| = \ldots\ldots$

c. $\left|\frac{1}{6}\right| = \ldots\ldots$

d. $-|9| = \ldots\ldots$

e. $\left|-\frac{6}{11}\right| = \ldots\ldots$

f. $|0| = \ldots\ldots$

g. $|-7| - |-7| = \ldots\ldots$

h. $|-3| \times |-2| = \ldots\ldots$

i. $|-40| \div |8| = \ldots\ldots$

j. $|0| + |-9| = \ldots\ldots$

k. $|-8| + |2| = \ldots\ldots$

l. $|-30| - |-10| = \ldots\ldots$

m. $|17| - |-17| = \ldots\ldots$

n. $|-49| \div |7| = \ldots\ldots$

o. $|-8| \times |-3| = \ldots\ldots$

p. $|0| \times |-1| = \ldots\ldots$

q. $|-30| \div |5| = \ldots\ldots$

2) Find the value of x:

a. $|x| = 10$

$x = \ldots\ldots$

b. $|x| = 8$

$x = \ldots\ldots$

c. $|x| = 0$

$x = \ldots\ldots$

d. $|7| = x$

$x = \ldots\ldots$

e. $|-3| = x$

$x = \ldots\ldots$

f. $|-19| = x$

$x = \ldots\ldots$

g. $|-1| = x$

$x = \ldots\ldots$

h. $-|6 \times 2| = x$

$x = \ldots\ldots$

i. $\left|-\frac{5}{9}\right| = x$

$x = \ldots\ldots$

3) Choose:

1. The absolute value of $\left(-5\frac{1}{6}\right)$ is ……..

a. $-5\frac{1}{6}$ **b.** $5\frac{1}{6}$ **c.** $-\frac{31}{6}$ **d.** 0

2. The opposite of $-\frac{3}{8}$ is ……..

a. $-\frac{3}{8}$ **b.** $\frac{3}{8}$ **c.** $-\left|\frac{3}{8}\right|$ **d.** $\frac{8}{3}$

3. All integers are also …….. numbers.

a. Counting **b. Natural** **c. Rational** **d. Even**

4. The rational number between -5.3 and -5.37 is …….

a. -5.30 **b.** -5.31 **c.** -5.36 **d.** 5.4

4) Put <, > or =:

a. $|-6|$ …….. $|-2|$

b. $|0|$ …….. $|-7|$

c. $|3.5|$ …….. $\left|-3\frac{1}{2}\right|$

d. $|-8.2|$ …….. 9

e. $|-9.54|$ …….. $|9.5|$

f. $5\frac{1}{3}$ …….. $\left|-\frac{16}{3}\right|$

Unit (2): Assessment

1) Choose:

1. The number −3 is a …….. number
a. Positive b. Negative c. Even d. Counting

2. Which of the following is a counting number?
a. 0 b. -5 c. 6 d. -7

3. The smallest non-negative integer is …….
a. 1 b. -1 c. 0 d. 10

4. The greatest negative integer is …….
a. 1 b. 0 c. -1 d. -10

5. The integer number consists of negative numbers and …….. numbers
a. Positive c. Counting
b. Natural d. Otherwise

6. An integer included between -4 and 2 is …….
a. -5 b. -1 c. 2 d. 3

7. The integer which comes just before -6 is ……..
a. -5 b. 7 c. -7 d. 5

8. The integer which comes just after -11 is ……...
a. -10 b. -12 c. 10 d. 12

9. The number of integers included between -2 and 4 is …….
a. 4 b. 5 c. 6 d. 3

10. The number of integers on the number line is …….
a. 1 b. 100 c. 0 d. Infinite

11.-5 is located to the right of the number …….. on the number line

a. -6 **b. 6** **c. -4** **d. 4**

12.The opposite of 0 is …….

a. 1 **b. -1** **c. 0** **d. -10**

13.The additive inverse of -12 is ……

a. -12 **b. 1** **c. 12** **d. 2**

14.All positive numbers are …… zero

a. Equal to **c. Greater than**

b. Less than **d. Otherwise**

15.The largest non-positive integer is ……..

a. 1 **b. -1** **c. 0** **d. -10**

16.The number -2.5 is a / an ……. Number

a. Counting **b. Integer** **c. Rational** **d. Natural**

17.The sum of any two opposite numbers is …….

a. 1 **b. 2** **c. 0** **d. -1**

18.The set of counting numbers …… the set of integer numbers

a. Belongs to **c. Rational**

b. Doesn't belong to **d. Natural**

Unit (3): Lesson (1&2)

Creating mathematical expressions

Analyzing mathematical expressions

1) Put a tick (√) to classify the following mathematical sentences into "Equation" or "Mathematical Expression" or "other":

Question	Equation	Mathematical Expression	Other
1) 25 × 17 = Y			
2) The difference of the two numbers is 13.8.			
3) 25.06 - 6.2 + 5 = Z			
4) Rana ran 10 km last week.			
5) 15 - Z = 18.5			
6) 37 × K = 500			
7) The sum of two numbers is 98.8			
8) 25.1 + 5.16			
9) 7.9 - 0.6 = 1.9 + 8.8			
10) Sally bought a skirt for 90.05 L.E. and a slipper for 400.9 L.E.			
11) 6.09 - 3 = N			

2) Complete the following table:

Expression	Number of Terms	Like Terms	Constants	Coefficients
1) $49 - x - 24.5$				
2) $4.2 + 1.5 + z$				
3) $0.7y + a + 63$				
4) $3x + 7x$				
5) $17 + 3 + y + z$				
6) $7 - 9$				
7) $2A + 9$				
8) $K + 5 + 6K$				
9) $9B$				
10) 7.8				
11) $5xyz$				
12) $-3y$				

Unit (3): Lesson (3)

Creating Algebraic & Verbal Expressions

1) Write verbal expression for each of the following algebraic expressions:

1. $a + 23$

_____
_____
_____

2. $5 - 7k$

_____
_____
_____

3. $5r + 12$

_____
_____
_____

4. $8(p + 9)$

_____
_____
_____

5. $y - 6$

_____
_____
_____

2) Write an algebraic expression for each of the following verbal expressions:

1. Subtract 9 from y.

_____
_____

2. **Two sevenths the number *k*.**

_____

_____

3. **Ahmed bought 9 packets, each of y pens.**

_____

_____

4. **The product of 9 and z.**

_____

_____

5. **The area of the square whose side length is x cm.**

_____

_____

6. **Divide the number L by 2.**

_____

_____

7. **9 less than triple a number k.**

_____

_____

8. **The product of a number and itself.**

_____

_____

9. **A number k increased by 15.**

_____

_____

10. **The product of 2x and 7.**

_____

_____

3) <u>Complete using the mathematical expression "$7x + 5y + 2x + 1$":</u>

a. The number of terms of a mathematical expression is ……..

b. Like terms are ……..

c. Coefficients are ……..

d. Constants are ……..

4) Alaa has 280 pounds, she saves y pound daily, after a week she paid half of the money as a donation. Write an algebraic expression for the remainder money with Alaa.

_____

_____

_____

_____

_____

5) Youssef runs two kilometers in 20 minutes. Write a mathematical expression that expresses the number of kilometers that Youssef runs in "t" minutes.

_____

_____

_____

_____

_____

6) A square of side length l, write an algebraic expression for the area of this square.

_____

_____

_____

_____

_____

Unit (3): Lesson (4,5&6)

Ordering of Operations & Exponents
Evaluating Algebraic Expressions
Applications on Algebraic Expressions

1) Complete:

a. **Five squared = ……..**
b. **7 to the power of 5 = ……..**
c. 2^2 **= ……..**
d. **3 cubed = ……..**
e. **p × p × p = ……..**
f. **Z × Z = -……..**
g. **5 × 5 × 5 = ……..**
h. **4 × 4 × 4 = ……..**

2) Use order of operations and exponents to simplify each of the following:

a. $((9-4)^2+4)+13+5^2$

_____

_____

b. $10+[10+(10-5)^2]+6$

_____

_____

c. $7+[5+(9-6)^2]-9$

_____

_____

d. $[(18-3)-(20\div 2)^2]\times 7^2$

_____

_____

e. $(20 \div 4)^2 + [(16 - 3) \times 5^2]$

_____

_____

3) Evaluate each algebraic expression for the given value of the variable:

a. $8s - 5$ **at** $s = 4$

_____

_____

b. $2b + 4$ **at** $b = 4$

_____

_____

c. $w + 2$ **at** $w = 8$

_____

_____

d. $\frac{c}{4} - 3$ **at** $c = 16$

_____

_____

4) The fare of a taxicab is 25 L.E. as a base fare and 12 L.E. for each km:

a. Write algebraic expression represent the total fare of the taxicab.

_____

_____

b. What is the fare for traveling 5 km.?

_____

_____

Unit (3): Lesson (7)

Determining Equivalent Algebraic Expressions

1) **If the two algebraic expressions $4n + 2n\,(n - 1)$ and $n(n + 1 + n + 1)$ are equivalent, find the value of the second expression for n = 1 in least number of steps.**

2) **Decide if the two algebraic expressions $3(x + 2)$ and $3x + 6$ are equivalent or not ?**

3) **Decide if the two algebraic expressions 2(x + 3) and 2x + 3 are equivalent and explain.**

Unit (4): Lesson (1)

Solving Algebraic Equations

1) Write verbal expression for each of the following algebraic expressions:

1. $x + 3 = 9$

_____

_____

2. $x + 7 = 20$

_____

_____

3. $x + 5 = 8$

_____

_____

4. $x + 9 = 9$

_____

_____

5. $x + 2 = 7$

_____

_____

6. $x - 5 = 1$

_____

_____

7. $x - 15 = 0$

_____

_____

8. $x - 50 = 40$

_____

_____

9. $x + 4 = 6$

10. $x - 5 = 0$

11. $x + 11 = 17$

12. $x - 22 = 28$

13. $5 + x = 8$

14. $1 + x = 1$

15. $7 - x = 0$

16. $10 - x = 3$

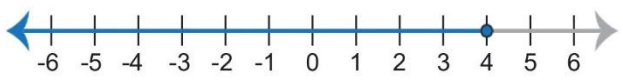
17. $6 + x = 6$

Unit (4): Lesson (2&3)

Solving & Exploring Inequalities

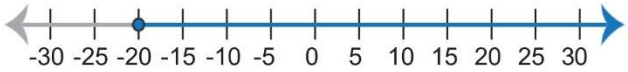
1) Write verbal expression for each of the following algebraic expressions:

1)



Inequality : _____

2)



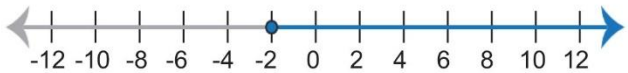
Inequality : _____

3)



Inequality : _____

4)



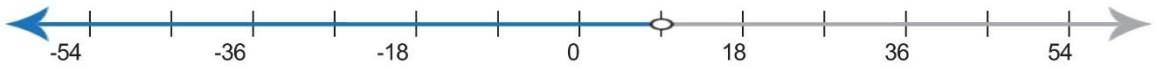
Inequality : _____

5)



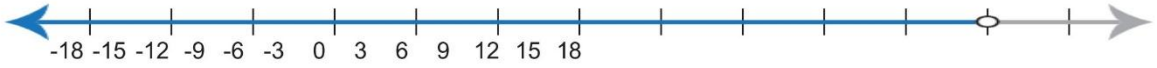
Inequality : _____

6)



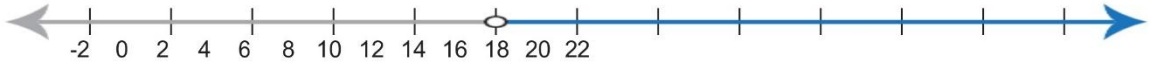
Inequality : _____

7)



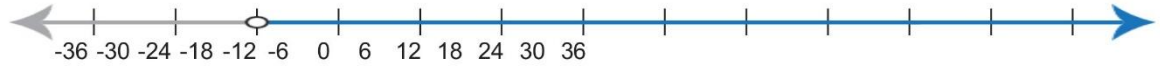
Inequality : _____

8)



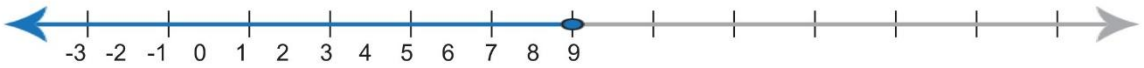
Inequality : _____

9)



Inequality : _____

10)

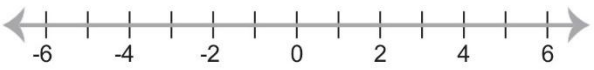


Inequality : _____

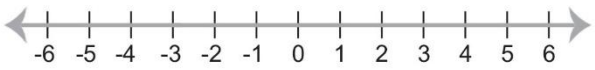
Graphing Inequalities

1) Graphing the inequality:

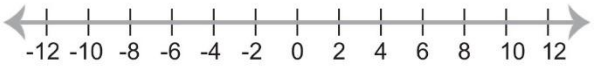
1) $x \geq -2$



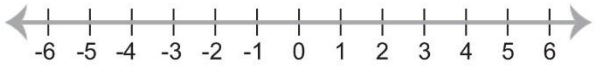
2) $x < 5$



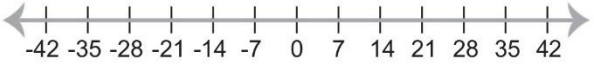
3) $x \leq 6$



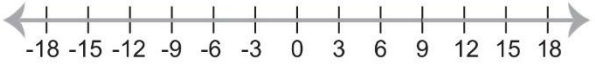
4) $x > 1$



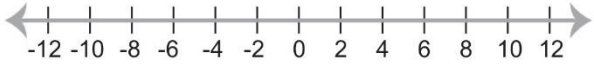
5) $x < -14$



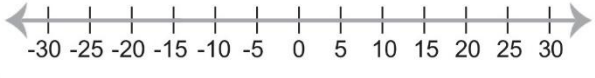
6) $x \leq 9$



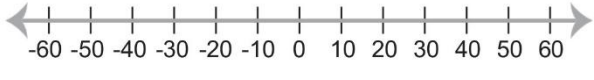
7) $x > -2$



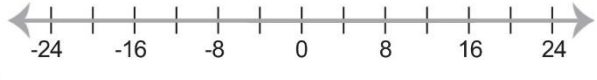
8) $x \geq -15$



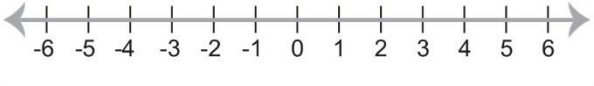
9) $x \leq -10$



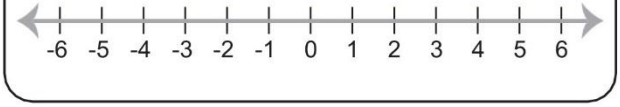
10) $x > 16$



11) $x > -1$



12) $x \geq -3$

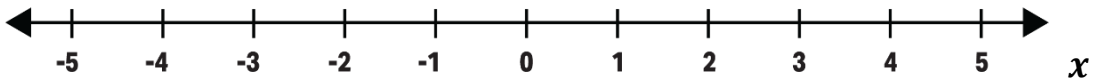


Exploring and Solving Inequalities

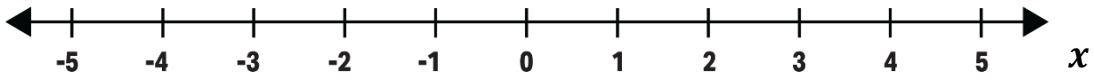
1) Match each inequality to the correct description:

1. $x > 3$	**i.** x **is less than or equal to 3**
2. $x \leq 3$	**ii.** x **is greater than or equal to 3**
3. $x \geq 3$	**iii.** x **is less than 3**
4. $x >< 3$	**iv.** x **is greater than 3**

2) Represent the inequality $x > 2$ on this number line:



3) Represent the inequality $x \leq 4$ on this number line:



4) Find 3 solutions of x in the set of integers:

a. $x \leq 3$

b. $x > -4$

c. $x > 0$

5) Find 3 solutions of x in the set of rational numbers:

a. $x \leq -6$

b. $x < -1$

c. $x \geq 1$

Unit (5): Lesson (1)

Dependent and Independent Variables

1) Determine the dependent and the independent variables:

a. The amount of food you eat and the weight you'll gain.

- **The dependent variable: ……**
- **The independent variable: ……**

b. The mark you get in Math exam and your right answers

- **The dependent variable: ……**
- **The independent variable: ……**

c. The amount of rain and the amount of growing plants.

- **The dependent variable: ……**
- **The independent variable: ……**

d.

Age (years)	**7**	**8**	**9**	**10**
Length (cm.)	**90**	**93**	**96**	**99**

- **The dependent variable: ……**
- **The independent variable: ……**

e. The spent hours in front of screens and the stance of sight.

- **The dependent variable: ……**
- **The independent variable: ……**

2) Notice then complete the table:

The algebraic equation	The independent variable	The dependent variable
f = r + 4		
w = 3u		
Y = 5X		
a = j - 7		
(E + 6) 2 = L		
4G = H		
k = 9 + 2z		
4S - 2 = Q		
X = 3L + 6		
5M - 3 = P		
b = n - 5		

Unit (5): Lesson (2)

Applications on Dependent and Independent Variables

1) The table shows the relationship between the minutes Vanessa spends walking, m, and the number of laps around the track she completes, c. Write an equation to represent this relationship.

c	*m*
3	18
5	30
6	36
8	48

a. Equation: ……

b. Dependent variable: ……..

c. Independent variable: ……..

2) Mohamed had to add 15 L.E. delivery to the cost of a cake.

a. The equation: ……

b. The dependent: …….....

c. The independent: …….

d. If X = 300 , then Y = …….

3) Ella's birthday party costs L.E. 30 for every guest she invites. Write an equation that shows the relationship between the guests, g, and the cost, c. Then complete the table.

g	*c*
5	
10	
15	
20	

a) Equation: ……

b) Dependent variable: ……..

c) Independent variable: ……..

Unit (5): Lesson (3)

The Relationship between Dependent & Independent Variables

1) Complete the table:

The rule	The equation
Multiply by 7	
Add 5	
Subtract 8	
Divide by 10	
Multiply by 3 and add 6	
Subtract 4 then multiply by 7	
Divide by 2 then subtract 1	
Add 8 then multiply by 5	
Multiply by 9 and subtract 3	

2) Complete the table:

The equation	The rule
$y = x = 4$	
$y = 3x$	
$y = 2x + 1$	
$y = x - 7$	
$y = (x + 6)2$	
$y = 9 + 2x$	
$y = 5x - 3$	
$y = -5$	

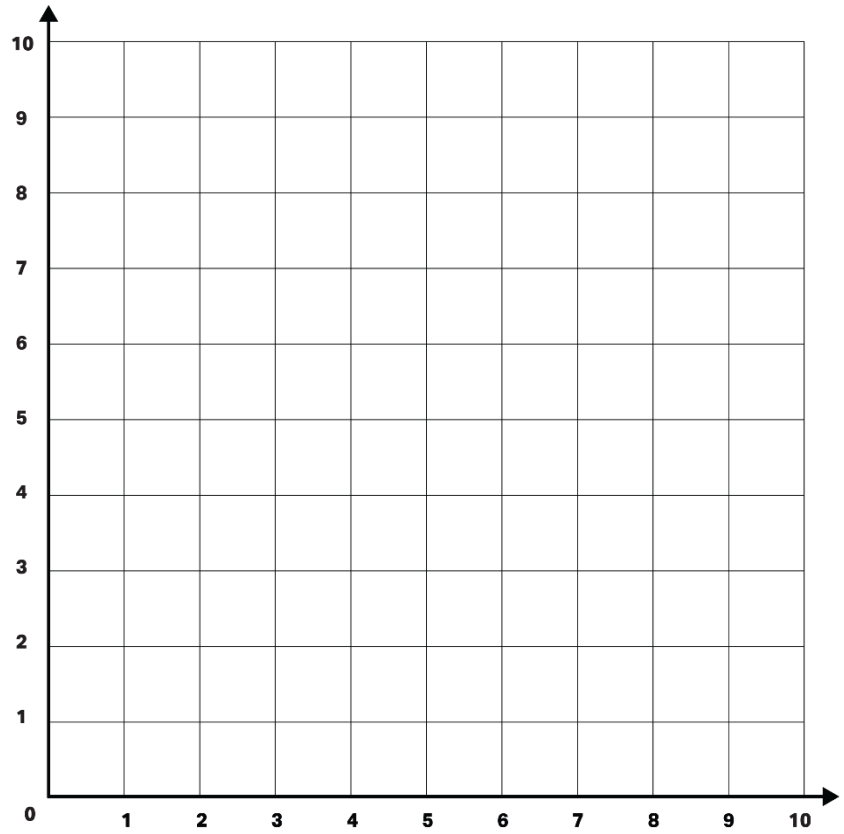
Unit (5): Lesson (4)

Graph Representation for Dependent and Independent Variables

1) Complete the tables according to the equation, then graph it:

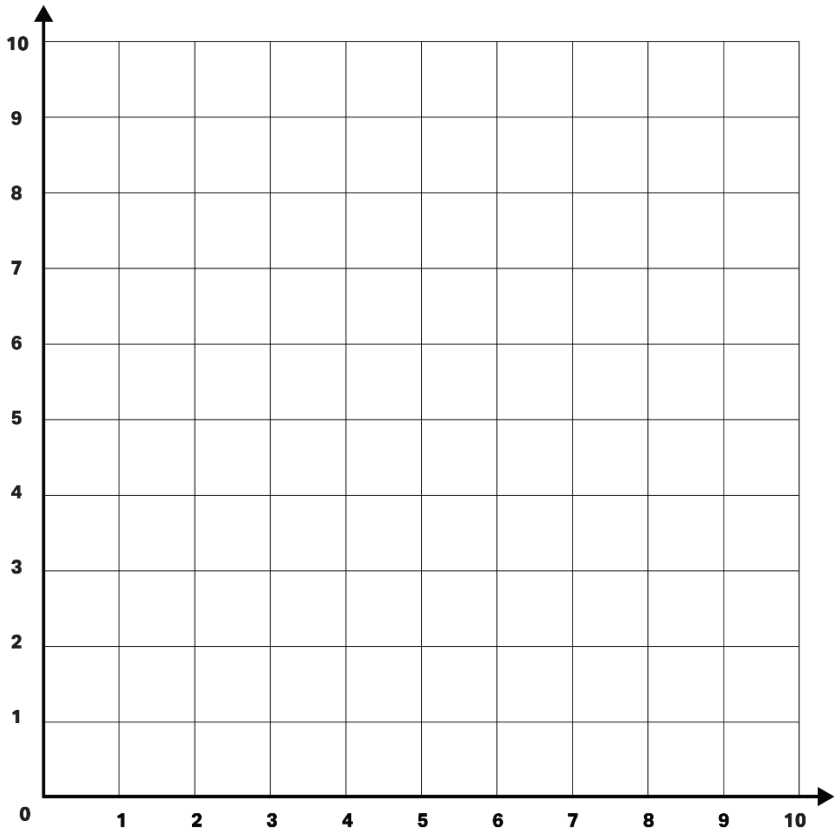
a. Equation : $y = 3 + x$

x	1	2	3	4
y				
(x, y)				



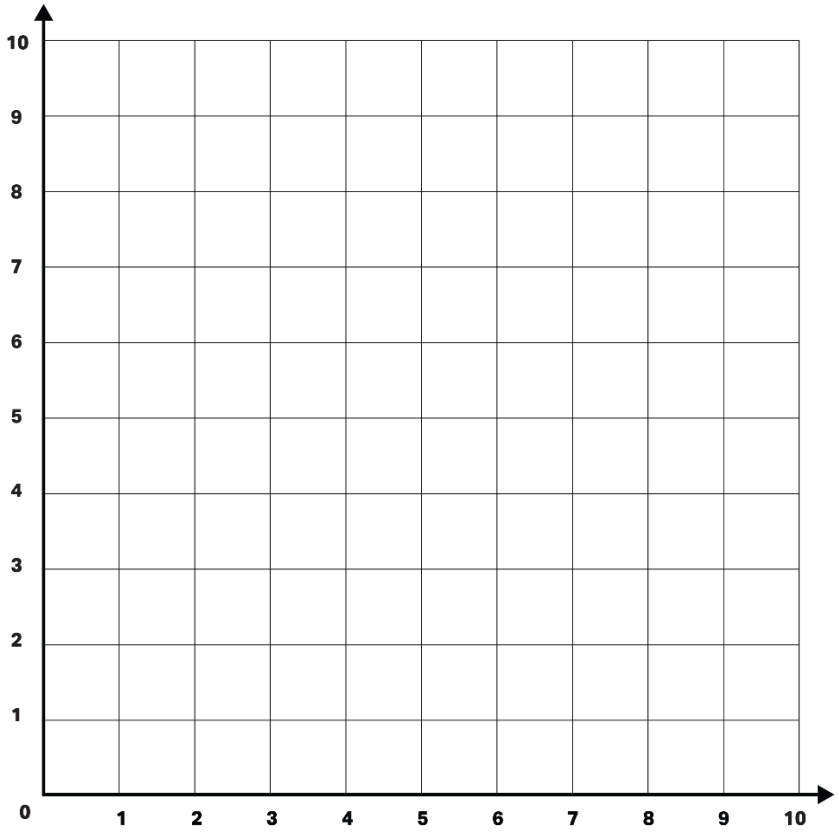
b. Equation : $y = x - 3$

x	3	5	7	9
y				
(x, y)				



c. Mera sells cakes and earns 2 L.E. for each cake

- **Equation: ……**
- ***x* describes: …….**
- ***y* describes: …….**



x	1	2	3	4
y				
(x, y)				

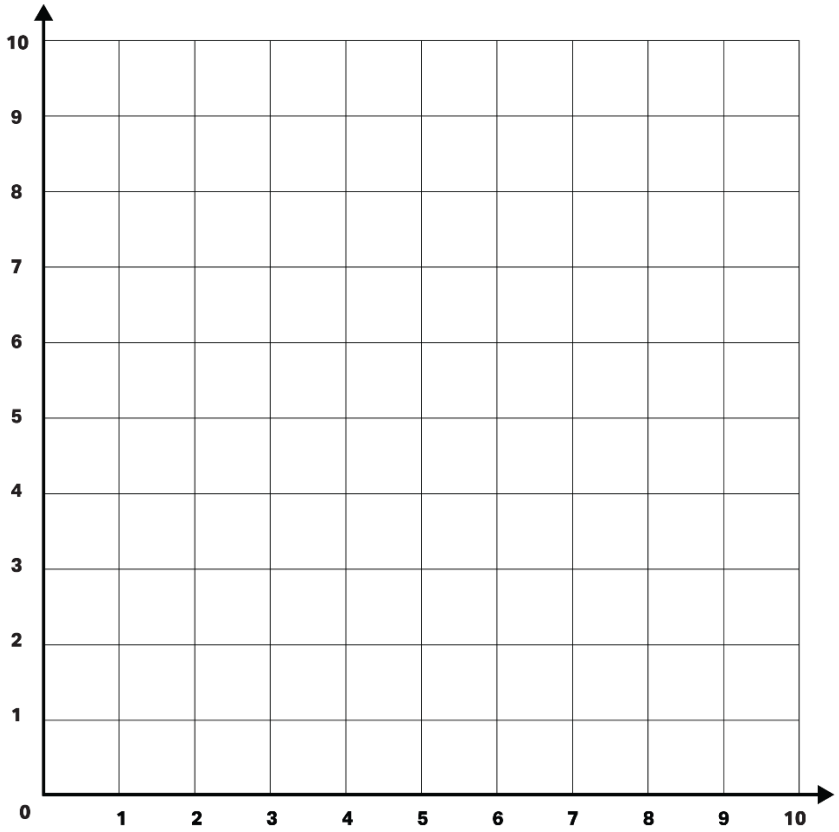
d. The rule: ……

Input x	1	2		4
Output y	4		12	16

- **Equation: ……**
- ***x* describes: …….**
- ***y* describes: …….**

e. Mera sells cakes and earns 2 L.E. for each cake

- **Equation: ……**
- **x describes: ……**
- **y describes: ……**



x	1	2	3	4
y				
(x, y)				

f. The rule: ……

Input x	12	20	8	4
Output y	7	11	5	3

- **The equation: ……**
- **If $x = 100$ then $y =$: ……**

Unit (6): Lesson (1)

Data and Statistical Questions

1) State whether each question below is a statistical question or not:

a. How old are you?
……
b. What is the favorite color for each student in your class?
……
c. Do you like fishing?
……
d. How many sandwiches does each one of your classmates have?
……
e. How many families live in your building?
……
f. What is your name?
……

2) State whether each question below is a statistical question or not:

a. How many students are in each class in grade 6? This statistical question is (numerical - categorial)
b. How old is each member of your team? This statistical question is (numerical - categorial)
c. What is the favorite movie for each one in your family? This statistical question is (numerical - categorial)
d. Which team does each member in your family cheer? This statistical question is (numerical - categorial)
e. What type of pets does every family own in your building? This statistical question is (numerical - categorial)
f. What is the favorite color for each member in your family? This statistical question is (numerical - categorial)
g. What is the height of each student in your class? This statistical question is (numerical - categorial)

3) Complete the following:

a. …… data is written in the form of numbers.

b. The number of letters in each students' first name is a …… data.

c. What is your favorite color? It is a ……question .

d. Categorical statistical question is answered in form of ……

e. What are the students' favorite colors in your class?
It is a …… question .

f. Types of statistical data are …… and …….

g. The number of pets owned by the students in your class, is …… data

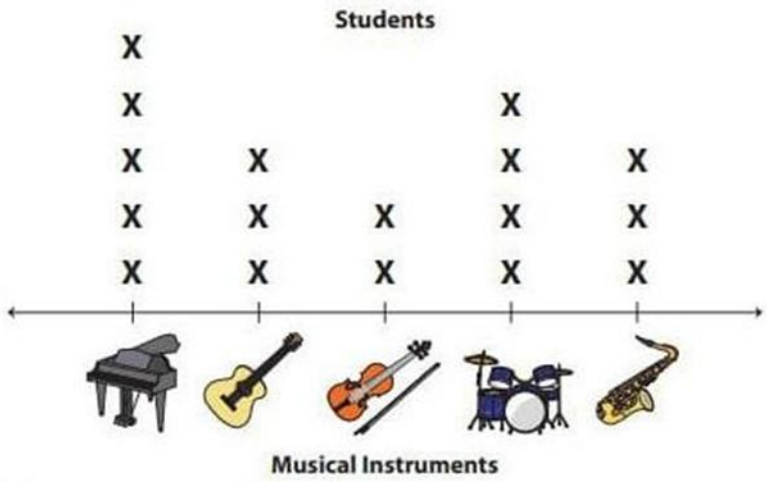
4) Choose the best graph in the following situations:

Questions	Dot plot	Bar graph	Histogram
How many students prefer the red color?			
How many students had grades in the exam from 15 to 20 ?			
What is the number of pupils in each class of the school?			
How many students in your class have a family of 5?			
How many trees in the garden have a height of 3 to 4 meters?			
What is the number of students who passed the Math exam?			

Unit (6): Lesson (2&3)

Exploring & Representing Histogram

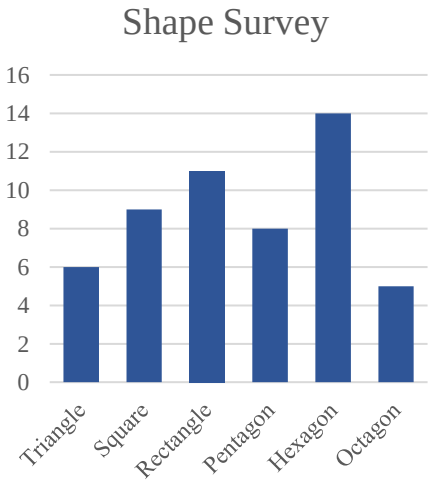
1) This dot plot shows the favorite musical instrument for some students:



X = 1 student

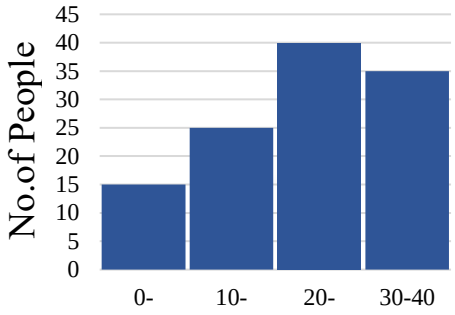
a. How many students were surveyed? .....
b. A suitable statistical question for this dot plot is .....
c. How many students like the guitar? .....
d. What is the most favored musical instrument? .....
e. How many students like the drums? .....
f. What is the least favored musical instrument? .....

2) A teacher conducted a survey about her students' favorite shapes, their answers are represented in the following bar graph:



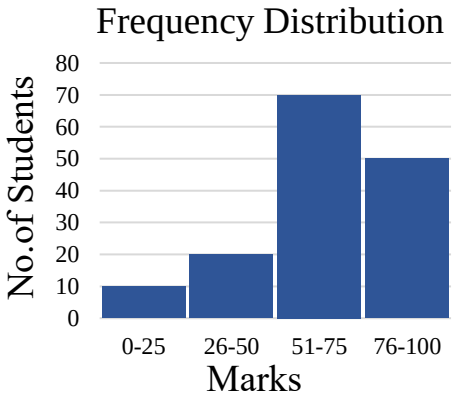
a. How many students were surveyed?
.....
b. What can be the label for the x-axis?
.....
c. What can be the label for the y-axis?
.....
d. How many students favored triangles?
.....
e. What is the least favorite shape among students?
.....

3) Answer according to this histogram :



a. A suitable statistical question for this histogram is ……
b. A suitable title for this histogram is ……
c. How many people were surveyed?
……
d. How many people are younger than 20 years old? ……
e. Which age group has the maximum number of people?
……
f. How many people are older than 10 years old?
……
g. How many people are aged 20 to 30 years old?
……

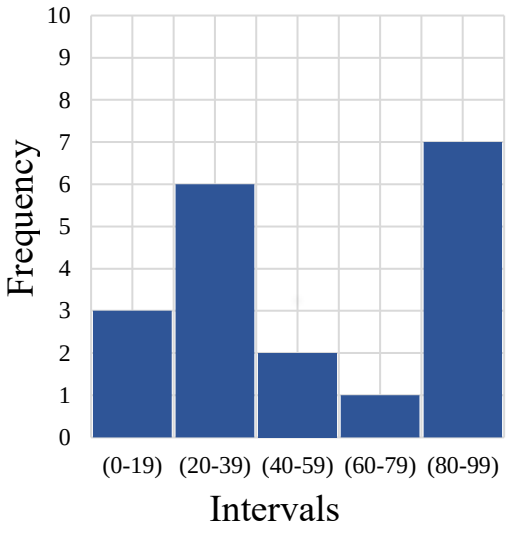
4) Answer according to this histogram :



a. What is the type of data represented in this graph? ……
b. A suitable statistical question for this bar graph is ……
c. How many students were surveyed?
……
d. What is the marks interval?
……
e. How many got marks between 26 & 75?
……
f. What are the highest obtained marks?
……
g. How many students got the highest marks?
……

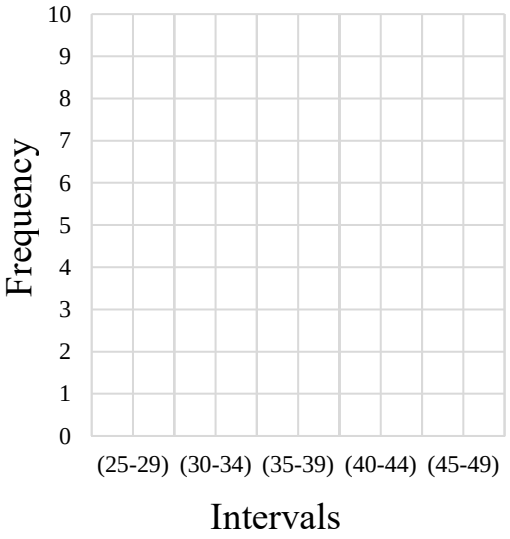
5) Use the following histogram to complete the following interval table:

Intervals	Frequency
0-19	……
20-39	……
40-59	……
60-79	……
80-99	……



6) Use the following histogram to complete the following interval table:

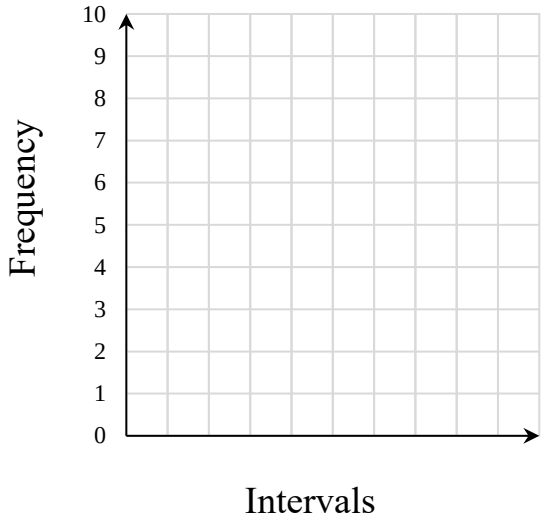
Intervals	Frequency
25-29	5
30-34	5
35-39	7
40-44	10
45-49	2



7) <u>The following frequency table shows the number of days of absence for 20 students in the class during the first term.</u>

Number of days	0	1	2	3	4	5	6	7	8
Frequency	3	2	3	4	2	1	0	3	2

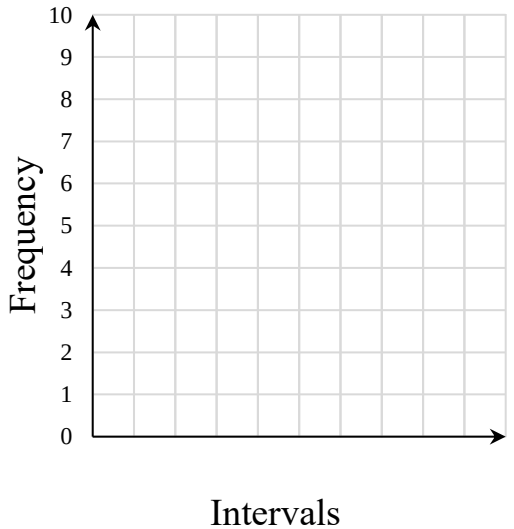
Intervals	Frequency
0-2	.....
3-5	.....
6-8	.....



8) <u>The following frequency table shows the number of days of absence for 20 students in the class during the first term.</u>

Scores	10	18	24	29	35	44	45	50	58
Frequency	2	6	5	9	6	10	8	3	1

Intervals	Frequency
10-19	.....
20-29	.....
30-39	.....
40-49	.....
50-59	.....

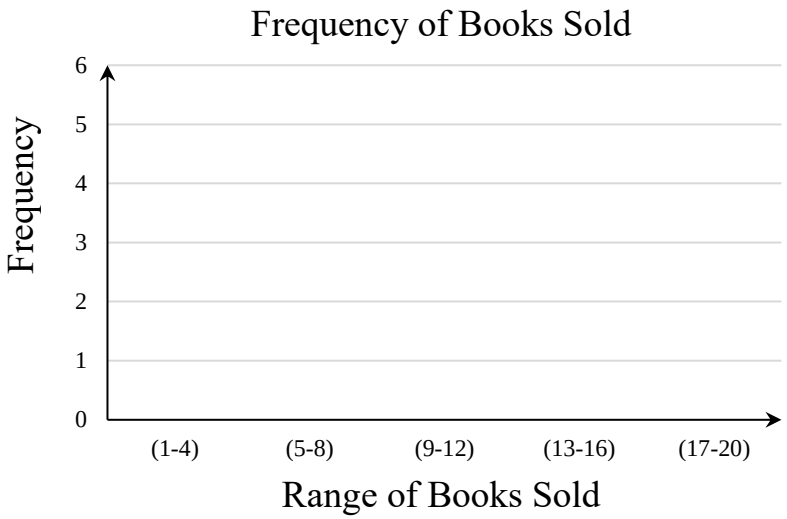


9) <u>Complete the frequency table then draw the histogram for each of the following:</u>

a. Ted goes to the market and sells books on 20 consecutive days. The sales results are shown below; he already organized them in order.

12, 7, 11, 6, 10, 18, 16, 5, 4, 14, 5, 7, 3, 17, 19, 20, 6, 15

Intervals	Frequency
1-4	.....
5-8	.....
9-12	.....
13-16	.....
17-20	.....

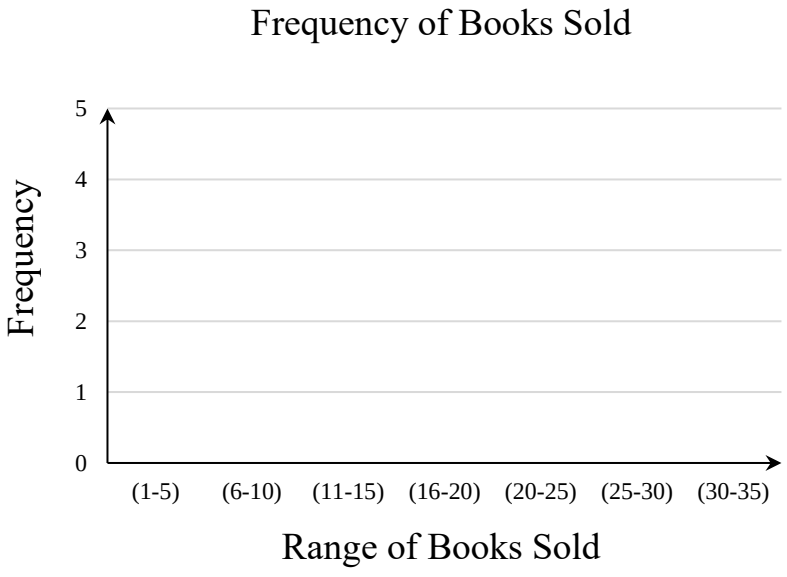


a. Mark takes part in a hockey tournament and scores many goals. The tournament consists of 20 matches. The number of goals that he scored are shown below. Organize them in order of interval 5 and take the help of the given table.

1, 11, 21, 20, 23, 25, 27, 31, 33, 8, 23, 3, 13, 24, 14, 15, 35, 9, 5, 15

Draw the graph and complete the frequency table

Intervals	Frequency
1-5	.....
6-10	.....
11-15	.....
16-20	.....
20-25	.....
25-30	.....
30-35	.....



Unit (6): Lesson (4)

Exploring & Representing Box Plot

1) Find the Median of the following data:

a. 5 , 7 , 8 , 3 , 6 :

the order: …… the median = ……

b. 9 , 3 , 7 , 5 , 6 , 3 :

the order: …… the median = ……

c. 7 , 8 , 5 , 3 , 7 :

the order: …… the median = ……

d. 7 , 6 , 2 , 9 , 6 , 0 , 6 :

the order: …… the median = ……

e. 2 , 0 , 5 , 8 , 11 , 6 , 5 , 7 :

the order: …… the median = ……

f. 3 , 9 , 1 , 8 , 2 , 3 , 6 :

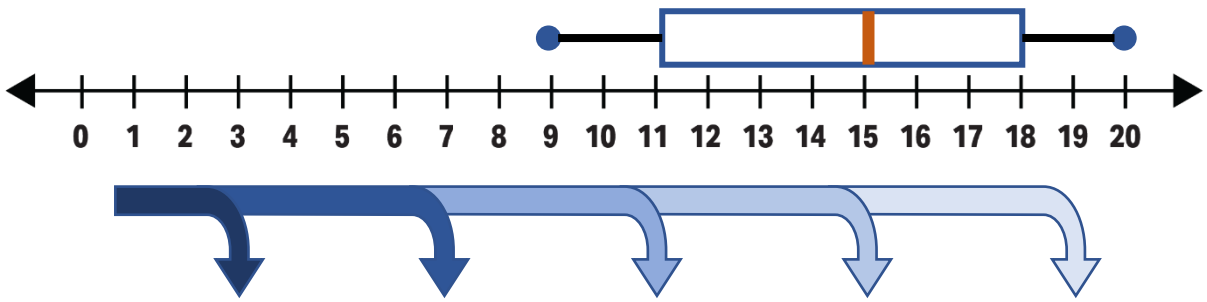
the order: …… the median = ……

g. 6 , 9 , 2 , 4 , 9 :

h. the order: …… the median = ……

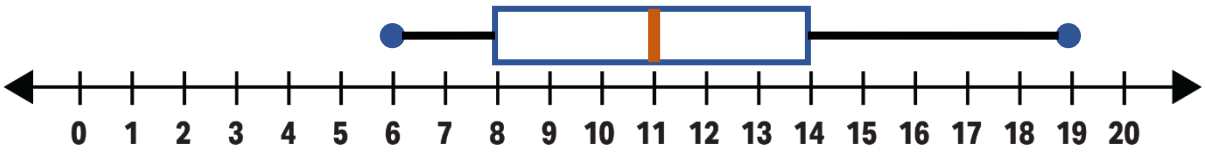
2) Complete each box plot of the following:

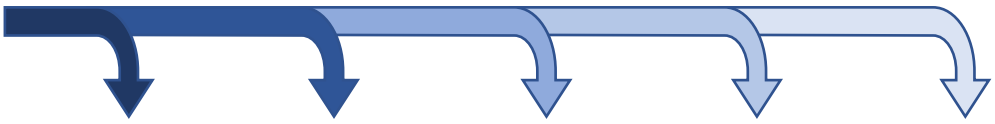
a.



Minimum Value	Lower Quarttile	Median	Upper Quartile	Maximum Value
……	……	……	……	……

b.

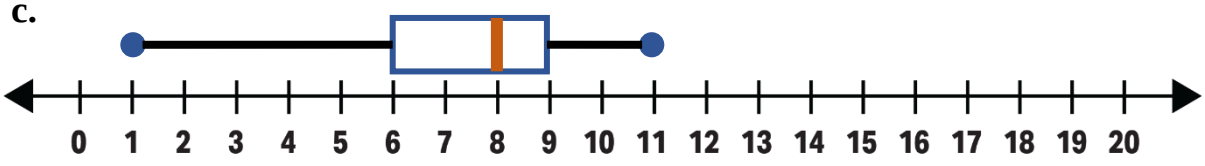


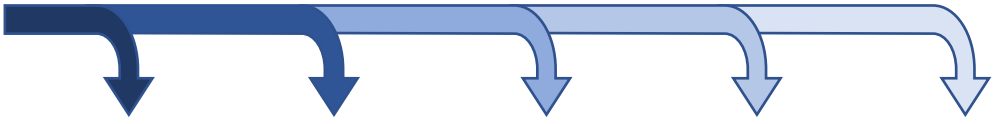


Minimum Value **Lower Quarttile** **Median** **Upper Quartile** **Maximum Value**

…….. …….. …….. …….. ……..

c.



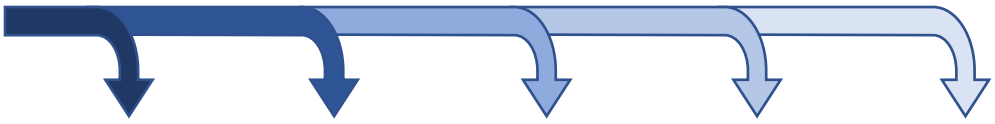


Minimum Value **Lower Quarttile** **Median** **Upper Quartile** **Maximum Value**

…….. …….. …….. …….. ……..

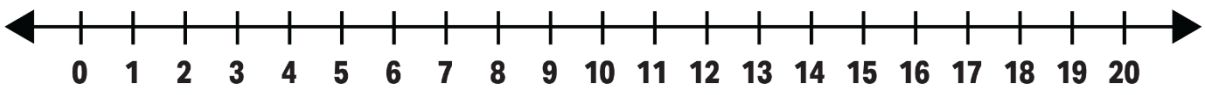
3) <u>Create a box plot to display the data:</u>

a. 9, 8, 3, 1, 10, 9, 5, 6, 5, 1. ……

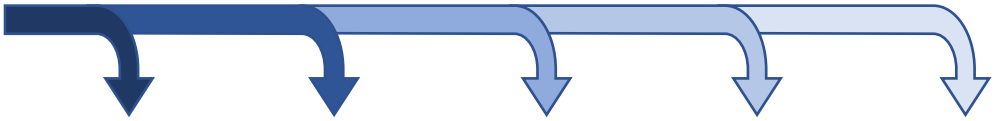


Minimum Value **Lower Quarttile** **Median** **Upper Quartile** **Maximum Value**

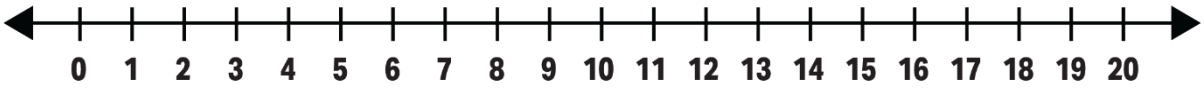
…….. …….. …….. …….. ……..



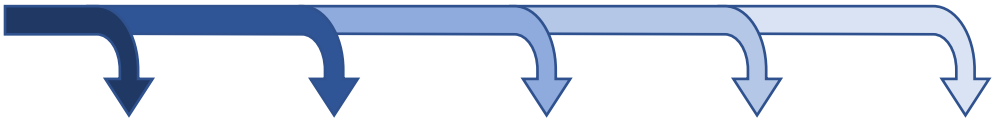
b. 15, 8, 9, 2, 7, 5, 4, 6. ……....



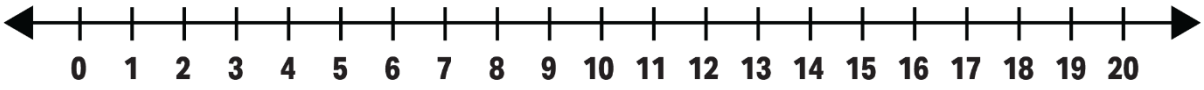
Minimum Value	Lower Quarttile	Median	Upper Quartile	Maximum Value
……..	……..	……..	……..	……..



c. 5, 6, 3, 7, 1, 0, 8, 8, 2, 1. ……



Minimum Value	Lower Quarttile	Median	Upper Quartile	Maximum Value
……..	……..	……..	……..	……..



4) <u>Write the 5-number summary for each set of data:</u>

1. 87, 14, 25, 35, 67, 73, 37, 39, 9

- **Minimum: ……**
- **Q1: ……**
- **Q2: ……**
- **Q3: ……**
- **Maximum: ……**

2. 48,5, 26, 89, 71, 36, 80, 92, 8, 16, 24

- **Minimum: ……**
- **Q1: ……**
- **Q2: ……**
- **Q3: ……**
- **Maximum: ……**

3. 39, 7, 27, 19, 49, 17, 13

- **Minimum: ……**
- **Q1: ……**
- **Q2: ……**
- **Q3: ……**
- **Maximum: ……**

4. 41, 52, 46, 34, 57, 32

- **Minimum: ……**
- **Q1: ……**
- **Q2: ……**
- **Q3: ……**
- **Maximum: ……**

5. 68, 29, 11, 23, 54, 31, 40, 62, 18, 44

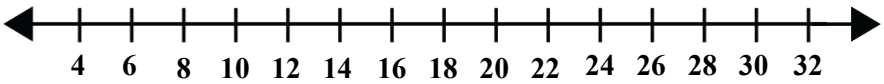
- **Minimum: ……**
- **Q1: ……**
- **Q2: ……**
- **Q3: ……**
- **Maximum: ……**

6. 75, 56, 70, 66, 45, 51, 72, 63

- **Minimum: ……**
- **Q1: ……**
- **Q2: ……**
- **Q3: ……**
- **Maximum: ……**

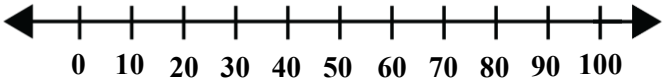
5) <u>Draw a box-plot for each set of data:</u>

a. 23, 10, 13, 30, 26, 8, 25, 18.



- **First Quartile =**
- **Second Quartile or Median =**
- **Third Quartile =**
- **Range =**

b. 35, 60, 20, 80, 95, 15, 40, 85, 75.



- **First Quartile =**
- **Second Quartile or Median =**
- **Third Quartile =**
- **Range =**

Unit (6): Lesson (5)

Applications on Data Representations

1) Choose the graph(s) which perfectly answers the following questions:

1. What is the most frequent value?

a. Histogram b. Box Plot c. Dot plot

2. What is the most frequent interval?

a. Histogram b. Box Plot c. Dot plot

3. What is the upper quartile?

a. Histogram b. Box Plot c. Dot plot

2) Write the type of graph that provides the best answer for each question (you may choose more than 1 answer):

a. Which graph shows group intervals?

……

b. How many people saved more than 25 thousand pounds?

……

c. Which graph shows grouped data?

……

d. How many students were surveyed?

……

e. Which graph shows the fewest number of books read?

……

f. Which graph shows summarized data?

……

g. How many players scored between 15 & 25 goals?

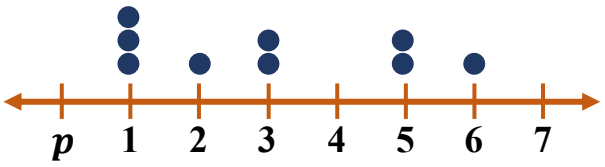
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Unit (7): Lesson (1&2)

Exploring the Balance of Data Sets , Interpreting Mean

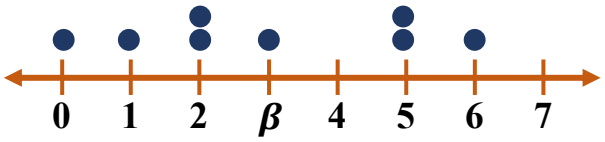
1) Determine the mean for the following:

a.



The Mean = …….

b.



The Mean = …….

c. 5, 7, 9, 11, 3, 7 **The Mean = …….**

d. 4, 9, 11, 8, 3 **The Mean = …….**

e. 18, 13, 11, 14, 9 **The Mean = …….**

f. If the number of goals scored by Al-Zamalek in 6 matches is :

4, 2, 3, 1, 0, 2. Calculate the mean for the number of goals.

……

Unit (7): Lesson (1&2)

Exploring Mean, Median, Mode & Outliers

1) Find the mode of each data set:

a. (9 , 5 , 5 , 9 , 7 , 3) The mode = ...

b. (4 , 9 , 1 , 0 , 8) The mode = ...

c. (10 , 19 , 17 , 16 , 15 , 17) The mode = ...

d. (1 , 2 , 8 , 9 , 1 , 3 , 1) The mode = ...

e. (3 , 0 , 8 , 2 , 9 , 30) The mode = ...

f. (3 , 3 , 5 , 3 , 6 , 6 , 6) The mode = ...

g. (red , blue , red , green , blue , blue)

The mode = ...

2) List the outliers in each of the following set of values:

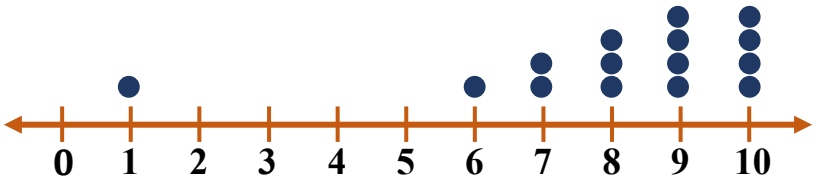
a. (6 , 8 , 2 , 8 ,7 , 5) The outlier = ...

b. (25 , 30 , 27 , 0 , 29) The outlier = ...

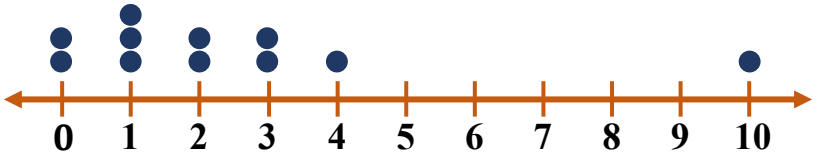
c. (24 , 24 , 200 , 25 , 26) The outlier = ...

d. (45 , 52 , 63 , 4 , 59) The outlier = ...

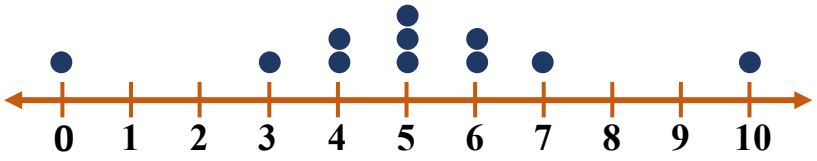
3) Choose the correct description: "Mean Increases, Mean Decreases, Mean Stays The Same"



.....

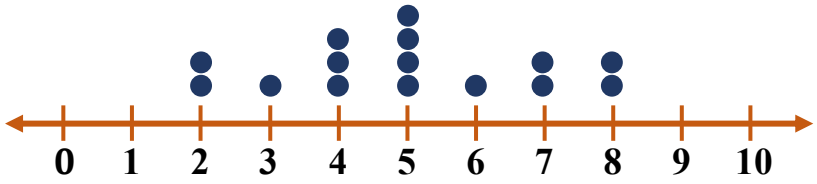


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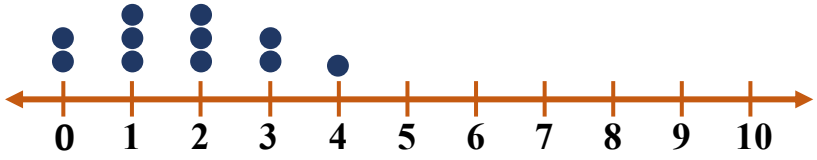


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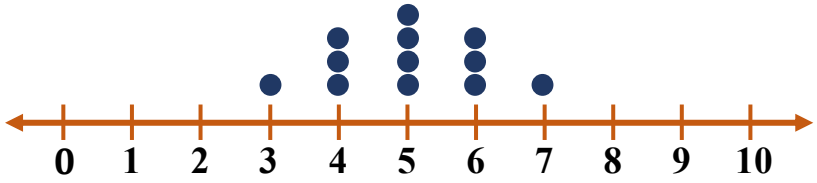
4) Choose the best measure for the central tendency: "Mean, Median, Either Mean or Median"



……



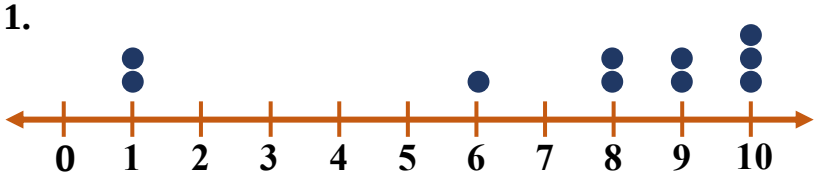
……



……

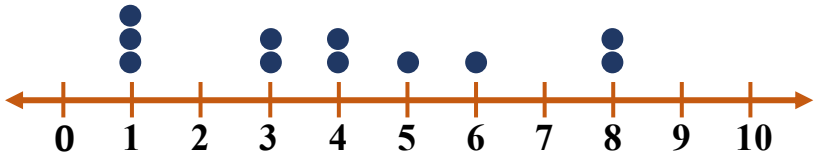
5) Find out for each of the following:

1.



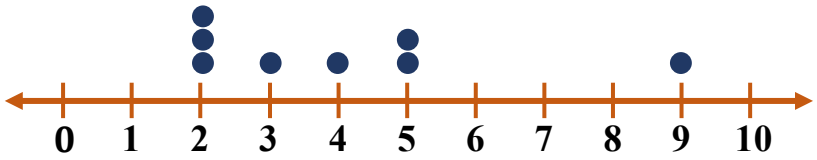
a. **The mean = ……..**
b. **The median = ……...**
c. **The mode = ……..**
d. **The outlier = ……....**

2.



a. **The mean = ……..**
b. **The median = ……...**
c. **The mode = ……..**
d. **The outlier = ……....**

3.



a. **The mean = ……..**
b. **The median = ……...**
c. **The mode = ……..**
d. **The outlier = ……....**

Unit (7): Lesson (4)

Exploring The Range

1) Find the range of each data set of the following:

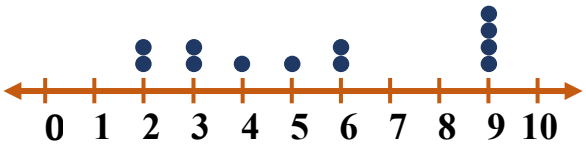
a. (45 , 25 , 13 , 30 , 35 , 5) **The Range = ……**

b. (6 , 2 , 7 , 7 , 5 , 3) **The Range = ……**

c. (9 , 2 , 7 , 6 , 3 , 4 , 9) **The Range = ……**

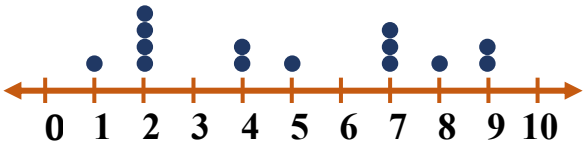
d. (7 , 9 , 2 , 7 , 2 , 7, 2 , 5) **The Range = ……**

e.



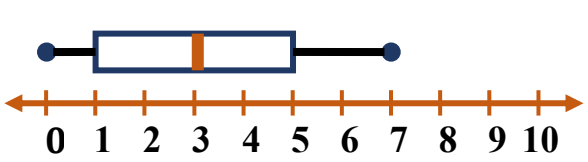
The Range = ……

f.



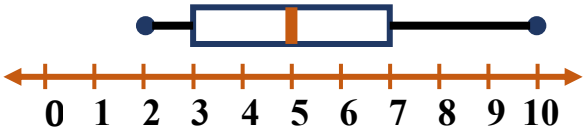
The Range = ……

g.



The Range = ……

h.



The Range = ……

i.

Name	Majed	Tawfik	Rami	Malik	Ahmed
Ages	15	11	18	13	14

The Range = ……

2) <u>Complete:</u>

a. (12 , 16 , 2 , 15 , 12 , 14 , 13)

The order: …….

The Range = …… The mode = ……

The median =…… The mean = ……...

The outlier = ……

b. (53 , 54 , 74 , 65 , 131 , 74 , 67)

The order: …….

The Range = …… The mode = ……

The median =…… The mean = ……...

The outlier = ……

c. (12 , 5 , 3 , 3 , 4 , 7, 8)

The order: …….

The Range = …… The mode = ……

The median =…… The mean = ……...

The outlier = ……

d. (1 , 10 , 7 , 3 , 14 , 9 , 12)

The order: …….

The Range = …… The mode = ……

The median =…… The mean = ……...

The outlier = ……

e. (5 , 7 , 13 , 11 , 2 , 1 , 2 , 14 , 16 , 10 , 3)

The order: …….

The Range = …… The mode = ……

The median =…… The mean = ……...

The outlier = ……

Revision(october's exam)

1) Choose the correct answer:

1) The constant for the algebric expression $\frac{a}{2} + 4$ is ……..

a) 0 b) 2 c)4 d)a

2) $36 + 45 = 9[\ldots\ldots + \ldots\ldots]$

a) 4, 7 b)4, 5 c)3, 5 d)7 5

3) The sum of 4 times m and 3 can be written as …….

a) $4m$ b)$4\,m + 3$ c)$3\,m + 4$ d)$4\,m - 3$

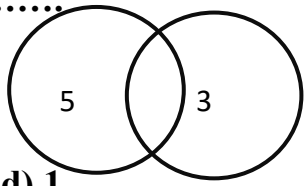
4) The greatest common factor of 9 and 27 is ……

a) 3 b)9 c)18 d)27

5) If we subtract 5 from the number x, The result is ……

a) $x - 5$ b) $5 - x$ c) $5\,x$ d) $x + 5$

6) From the opposite venn diagram, the LCM is …….



a) 0 b) 7 c) 15 d) 1

7) The coomon factor of all numbers is ……

a) 0 b) 2 c) 3 d) 1

8) The mathematical eexpression $4m - 7n + 5$ contains …….terms

a) 3 b) 4 c) 8 d) 1

9) The LCM of 10 and 15 is ……..

a) 6 b) 30 c) 9 d) 18

10)…….. is a multiple of any number

a) 0 b) 1 c) 2 d) 3

11) 540÷15 =……

a) 32 b) 50 c) 36 d) 34

12)The distance between 5 and $|-9|$ on the number line is ……. Units

a) 10 b) 0 c) -5 d) 4

13) Which of the following is the smallest number ?.....

a) -3.8 b) -8.3 c) -2.3 d) -3.2

14) The constant in the mathematical expression is $2x + 5y + 3$ ……

a) 2 b) 3 c) 5 d) none

15) $|-2.27| <$……

a) 2 b) 2.26 c) 2.28 d) -3

16) $1\frac{1}{6} - \frac{1}{2} =$ ……

a) $1\frac{1}{3}$ b) $\frac{2}{3}$ c) $\frac{1}{2}$ d) $\frac{1}{4}$

17) $18 + 45 = 9$[……..]

a) 9+3 b)2+5 c) 9-3 d) 4+2

18)$|4| \times |-4| =$…….

a) -64 b)-16 c)0 d)16

19)The absolute value of $-3\frac{1}{4}$=……

a) $3\frac{1}{4}$ b) 0 c) $-3\frac{1}{4}$ d) $-\frac{13}{4}$

20)The opposite of $-\frac{3}{7}$ is……..

a) $-\frac{3}{7}$ b) $\frac{3}{7}$ c)$-\left|\frac{3}{7}\right|$ d) $\frac{7}{3}$

21) G.C.F of 5 & 12 is ……..

a) 0 b) 1 c) 7 d) 14

22) $\frac{1}{3}+\frac{1}{5}$ =……

a) $\frac{2}{9}$ b) $\frac{9}{5}$ c) $\frac{8}{15}$ d) $\frac{1}{20}$

23) -23.5 is a …….

a) rational number b) integer

c) counting number d) natural numer

2) <u>Complete the following :</u>

1) $318\div 6$ =……
2) $\frac{4}{5}+\frac{2}{3}$ =……
3) $18+45$ =…….×[……+……..]
4) The opposite of 0 is ……..
5) All the prime nubers are odd except ……..
6) $5\times(3+6)$=(…….×…….)+(……×……)
7) $2\frac{2}{3}-1\frac{5}{24}$ =……
8) $4928\div 14$ =……
9) If $|x|=15$, then x =……
10) 8 decreased by three times a number is ……
11) The absolute value of $-3\frac{1}{4}$ =…….
12) The quotient of a number h by 9 ……

13) **The opposite of $-\frac{4}{9}$ is …….**

14) **The constant in the algebric expression $7b - 3.2$ is …….**

15) **5 meters below the sea level …… (write an integer)**

16) **The integers between -3 and 1 are …….**

17) **The mathematical expression which represents "6 increased by double a number" is …….**

18) **The number $1\frac{1}{5}$ in the form of $\frac{a}{b}$ is ……**

19) **The like terms in the expression $3a - 4b + 9a - 2$ are …….**

20) **The opposite of $-|-5|$ is ……..**

21) **The number of terms of the algebric expression 5x-7xy+1**

Is …….

22) **A square of side length "S" cm has a perimeter of …….cm.**

23) **$-1, 3\frac{3}{5}, |-8|, 2, \left|-2\frac{3}{5}\right|$ in an ascending order is**

……

3) **<u>Essay questions : Answer the following</u>**

1) **A fruit seller wants to divide 1485 oranges among 55 boxes, how many oranges are there in each box ?**

……
……
……
……
……

2) **Dina divided 24 apples and 27 oranges into boxes so that the boxes contained the same number of oranges and the same number of apples. What is the greatest number of boxes she can make? Write a numerical expression for the sotuation. (use venn diagram)**

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……

3) If the ticket of a park is 120 L.E. and fees of 50 L.E. for each game you play inside.

a) Write an algebric expression to present the relation between the number of games you played and the total cost.

……

b) How much money you will pay in total if you want to play 10 games?

……

4) Write verbal expression for each of the following algebric expressions :

a) $x - 6$

……..……..….

b) $2y$

……..……

c) $k - 2$

……..……

d) $2n - 1$

……..……

e) $2m - 3$

……..……

f) $3(m + 7)$

……..……

g) $\frac{1}{2}b + 15$

……..……

5) Write an algebric expression for each of the following verbal expressions:

a) A number less than ten

……..……

b) Subtract 3 from, 5 times a number

……..……

c) Distribute $(x + 2)$ piece of candy among 4 boys.

……..……

d) 5 less than twice a number m

……..……

e) The product of a number and 7

……..……

f) The sum of 2 times x and 5

……..……

g) Twelve less than a number

……..……

4) <u>Choose the correct answer:</u>

1) In the equation $384 \div 16 = 24$, the dividend is ……..
a) 384 b) 16 c) 24 d) 0

2) If $40 \div 5 = 8$, then the remainder is ……
a) 40 b) 5 c) 8 d) 0

3) If mona has 17 oranges and she wants to distribute them equally among 3 of her friends, how many oranges are left?
a) 17 b) 3 c) 5 d) 2

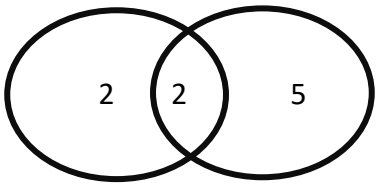
4) Which is the correct relation which represents the following statement: (distribute 16 crayons equally among 4 students)

a) 16×4 b) $16 + 4$ c) $16 \div 4$ d) 16 – 4

5) From the opposite venn diagram the G.C.F = ……

a) 6 b) 5

c) 2 d) 30



6) Which of the following numbers is a counting number?

a) -5 b) 2 c) -2 d) 0

7) The integer numbers consists of negative numbers and …… numbers

a) Positive b) counting c) natural d) otherwise

8) Which integer represents the following :

a) 2 b) 20 c) -20 d) 2

9) An integer included between -2 and -3 is ……..

a) 5 b) -3 c) -4 d) -1

10) The integer which comes just before the number -5 is …….

a) -4 b) -6 c) 4 d) 6

11) 3.2 …….. the set of counting numbers

a) Belong to b) doesn't belong to c) subset of d) not sunset of

12) $|-5| >$…….

a) $|-4|$ b) $|6|$ c) 9 d) 5

13) Which of the following is an algebric expression?

a) 3+5 b) 9+8-0 c) 2m+5 d) 3+(2+1)

14) The number of terms in the algebric expression x+y-5 is…….

a) 1 b) 2 c) 9 d) 4

15) Which of the following are like terms ?

a) $3x, 3$ b) x, y c) $x, 2x$ d) $3m, 3n$

16) The coefficient in the algebric expression $3x + y = 2$ is ……..
a) 3, 1, 2 b) 3, 1 c) 3 d) 2

17) Subtracting 5 from double of a number =……..
a) 5-b b) 2b-5 c) 5-2b d) b-5

18) The algebric expression of "three times a number added to 14" is
a) $3x + 14$ b)$3x - 14$ c) $x + 14$ d) $x - 14$

19) Twice the sum of a number and 3 is ……
a) $2y + 3$ b) 2(y+3) c) $2y - 3$ d) 2(y-3)

20) Carla saved x L.E and her father gave her 10 L.E, she wll have
a) $10x$ b) $x - 10$ c) $10 - x$ d) $x + 10$

21) Adam is x ears old. How old was he before 5 years ?
a) $5x$ b) $5 - x$ c) $x - 5$ d)$5 + 10$

5) <u>Complete the following:</u>

1) The quotient in the equation 438÷5=87 R3 is……
2) The common factor of all numbers is ……
3) 6(2+4)=12+……..
4) The G.C.F of 3 and 5 is ……..
5) The opposite of zero is……
6) The smallest counting number is ……
7) The greatest negative integer is ……..
8) The smallest non-negative integer is ……..
9) The opposite of $|-2| =$……
10) If $|x| = 5$, then x= ……or……..
11) $|-2| \times |2| =$……..
12) $|5| + |0| =$…….
13) The number of terms in the algebric expression $5x + 3$ is ……..
14) The constant in the expression $3x + 3y + 4$ is …….
15) The coefficient in the algebric expression 5m-3
16) The algebric expression of "the quotient of a number x by 2" is ……..
17) The algebric expression of "twice the sum of a number and 5 " is…….
18) The verbal expression of "$4 + x$" is …….
19) The verbal expression of "2m-5" is ……..

6) <u>Answer the following:</u>

1) A factory produces 945 bottles of juice weekly. How many bottles did the factory produce daily?

2) …….

3) ……...……..

4) …….

5) …….

6) Use venn diagram to find: G.C.F and L.C.M of 12 and 18

…….

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…….

7) Ahmed has 8 oranges and 12 bananas, He wants to make baskets with the same number of each fruit in each basket

a. What is the greatest number of baskets he can make?

b. Write the expression which represents that.

…….

……...……..

…….

…….

8) Write the algebric expression which represent: Subtract 5 from the half of a number

…….

……...……..

…….

…….

9) Write the verbal expression which represent: 3(n+5)

…….

10) Arrange the following numbers from the smallest : -5, -12, 23, $|-3|$ and $-|-12|$

……

Revision (november's exam)

1) Complete the following:

1) Th coefficient of the algebric term 5ab is ……..
2) The sum of two integers is S and one of them is 10, then the other number is ……..
3) Number of like terms in "3n+3+2n" …….
4) Twice a number added to 5=……..
5) The value of $(14\text{-}x^3) \div 2$ if $x = 2$ is ……
6) 5 Squared added to 6 cubed is …….
7) The dependent variable in the "a=3b' is …….
8) The equation that expresses the relation add 5.1" is ……
9) If $y = \frac{1}{2}x + 4$, $x = 6$, then y =……
10) $6^2 \div 3^2 \times 2$ =……..
11) if $x + 9 = 13$, then $7x$ =……
12) if $5y = 45$,then $y - 2$ =……..
13) 10 less a number is written as …….
14) In 6^5 the base is ……. The exponent is ……
15) (……..,……)is the origin point
16) "y is 7 times x added to 6" is written as ……..
17) The inequality that represents the megative numbers ……
18) The smallest solutiob in the inequality $x \geq -1$ is …….
19) If $\frac{x}{5} = 4$, then $3x$ =……..
20) In the equation "8a=b" the independent variable is……
21) If $\frac{x}{3.1} = 2.4$, then $x + 2$ =……..
22) If $x - 13 = 15$ then $\frac{x}{2}$ =……

23) The verbal phrase for 2n+8=m is …….
24) The inequality thatbrepresent non-positive numbers is……
25) The constant in the expression $5x + 3$ is …….
26) "subtract 5 from three times a number" is written as ……..
27) Twice the sum of a numbber and 3 is written as …….
28) If the age of mona now is X then her age after 5 years is ……
29) Three times a number less than 6 is ……..
30) The verbal phrase of $2(x - 5)$ is …….
31) The value of the expression $5x + 2$ when $x = 2.1$ is ……..

32) $9 \times 9 \times 9 \times 9 \times 9 \times 9 = 9\ldots$

33) $8 - 3 \times 2 \div (4 - 3)^2 = \ldots\ldots\ldots\ldots\ldots\ldots$

34) The first operation in the expression $5(6 - 3)^2$ is ……

35) Number of terms in the expression "$2x + 3 - 5$" is ……

36) Samar has 50 L.E she brought 3 pens each for x L.E , then the left money with her is ……

37) The value of $5(2h - 3) + 1$ at h=2.5 is ……..

38) $5x^2 + 8 \div (6 - 4) \div 2$ at $x = 3$ is ……

39) Write 3 solutions of the inequality $x < -2$……

40) A number is more than equal to 10 can be written as …….

41) "m equal the product of n and 3" in equation is ……

42) The dependent variable in the equation "$x + 3 = y$" is ……

43) In the equation $y = 2x$ the value of y when $x = 2.1$ is …….

44) (5, ……) satisfies the equation $y = 2x + 1$

45) Is the outut of the equation $y = x + 1$ is 2 then the input is

46) If $y = \frac{1}{3}x - 5$, $x = 18$, then $y = \ldots\ldots\ldots\ldots\ldots$

47) The inequality that represent all values less than -6 is …….

48) $2^3 + 2^3 = \ldots\ldots\ldots\ldots$

49) If the price of 8 pens is x, then the price of each one is ……..

50) $1^5 = \ldots\ldots\ldots\ldots$

51) Salah saves Z pounds per day, How much he saves in a week?.....

52) If the base is 5 and the exponent is 3 then the exponential form is …….

53) In the expression ($2x$+1) and ($5x$-4) if we put $x = 5$ then they will be equivalent ot not?

54) Twice of subtracting 5 from W is written as …….

55) Sara is now 25 years old, how old was she h years ago?

2) Choose the correct answer:

1) If $y = 7 + 2x$, then (……,11) satisfies the rule
a) 1 b) 7 c) 2 d) 4

2) The ordered pair that satisfies the ruke ($y = 3x - 2$)
a) (0, 2) b)(2, 4) c) (3, 4) d) (1, 0)

3) All the following are solutions of the inequality $x > 2$ except
a) 5 b) 6 c) 3 d) 2

4) Mohamed has 45 L.E, his friend has less money than mihemed, thn he may has……..L.E
a) 45 b) 40 c) 89 d) 90

5) A number if added to 17 the sum is 28, then the number is …….
a) 18 b) 45 c) 11 d) 17

6) If $x + x + x + x = 20$, then $x =$
a) 20 b) 4 c) 5 d) 12

7) If the product of number f and 8 is 64, then f=……
a) 9 b) 7 c) 8 d) 6

8) The volume of the cube whose edge length is 6 is …….cm^2
a) 36 b) 6 c) 6^3 d) 215

9) The area of square whose side length =5 cmm is …….cm^2
a) 15 b) 5 c) 5^2 d) 20

10) Take away twice the number K from 15 written as
a) 2K-15 b) 15+2K c) 15-2K d) 15-K^2

1) **Find the value of exprssion 8+2(6-2)÷ 2^3**

……

2) **If the ticket of entering a car park is 25 L.E and 10 L.E for each hour you spend. Write an expression to present the relation between the total cost and number of hours**

……

……

And what is the cost of spending 5 hours in the park?

3) **Write an equation use the variables x and y, where x is the independent m write the equation "multiply by 8 and add 3" m substitue $x = \frac{1}{4}$ to evaluate y**

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3) **<u>Solve the following equations:</u>**

1) $8 + x = 21$

……

……

……

2) $3y = 18$

……

……

……

3) $z - 6 = 19$

……

……

……

4) $\frac{f}{2} = 10$

……

……

……

4) Graph each of the following inequalities in N and Z:

1) $3x > 15$

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……
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……
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……

2) $x - 4 \leq 10$

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……

3) $\frac{y}{3} > 8$

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5) Find the solution set of the inequality $6 - x < 4$ if the substitution set is {0, 1, 2, 4}

……
……
……
……
……
……

6) Graph the following inequalities $x > 3$ and $x < -3$ on the number line, then answer with "yes" or "no" :

1) **They have no points in common**
2) **Does 3 belong to the solution set of each of them**
3) **Does -3 belong to the solution set of only one of them**
4) **The inequality $x > 3$ includes all the values to the right of 3 on the number line**
5) **The inequality $x < -3$ includesn all values to the right of -3 on the number line**

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7) Find the value of the algebric expression in each of the following:

1) $34 + b(4^2 \div 2)$ when b=6

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……
……
……

2) $9 + 3(m^2 + 2)$ for m =5

……
……
……
……
……

8) Follow th order of performing operations, then find the value of each of the following:

1) $[5^2 \times (6^2 \div 9)] - 24$

……
……
……
……
……

2) $4^2 + (15 - 7) \times 2$

……
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……
……
……

9) Evaluate each of these algebric expression using two positiveintegers of your choice. And determine if they are equivalent or not.

1) $9(x + 6)$ **&** $9x + 6$

……
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……
……

2) $3(2x + 1)$ **&** $6x + 3$

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……
……
……
……

3) $4x + 6$ and & $3(x + 2)$

……
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……
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……

10) A worker in a factory receives a daily wage of 430 pounds for working for a specified number of hours, in addition to an amount of 100 pounds for every additional hour of work.

1) **Write an algebric expression that expresses the wage that the worker recieves on a day on which he worked for y of overtime hours.**

……
……
……
……

2) **If the number of overtime hours is 3 hours, what is the amount that he gets for that day?**

……
……
……
……

Final Revision

1) Choose the correct answer:

1) The mode of values:5, 3, 2, 5, 8, 1, 5 and 4 is…….
a) 1 b) 2 c) 3 d) 5

2) The number of terms in the following expression: $6x+1$ is ……
a) 2 b) 1 c) 3 d) 0

3) The otlier of the following values: 32, 37, 36, 5 and 40 is ……
a) 32 b) 40 c) 5 d) 4

4) In the equation: m=3n+4, the dependent variable is ……
a) m b) 3 c) n d) 4

5) The median for the set of values: 114, 120, 97, 104, 127 is …….
a) 97 b) 120 c) 114 d) 104

6) The value of the expression: $5n-2$ for n=1 is ……..
a) 5 b) 3 c) -2 d) 1

7) The additive invers of -2 is ……..
a) -2 b) 2 c) 0 d) 4

8) The smallest counting number is …….
a) -1 b) 0 c) 1 d) 2

9) 16+24=8(2+…….)
a) 24 b) 16 c) 2 d) 3

10) In the equation: 1860÷32 =58 R4, the remainder is …….
a) 1860 b) 32 c) 58 d) 4

11) In the equation 5t=20 the value of t=…….
a) 3 b) 4 c) 5 d) 6

12) $-\frac{1}{2}$ ……..zero

a) $<$ b) $>$ c) = d) $\leq$

13) The mean of the values: 3, 5, 4, 7 and 6 is …….

a) 3 b) 4 c) 5 d) 6

14) The smallest non negative integer is …….

a) -1 b) 0 c) 1 d) 2

15) 4……0

a) $>$ b) $<$ c) =

16) if y=7+3x, then (……., 10) satisfies the equation.

a) 1 b) 0 c) 2 d) 4

17) The G.C.F of 10 and 7 is ……..

a) 2 b) 18 c) 40 d) 80

18) The smallest solution of inequality $x < -2$ is ……..

a) -3 b) -1 c) 0 d) -2

19) 10 - $3\frac{1}{4}$ = ……..

a) $7\frac{1}{4}$ b) $6\frac{1}{4}$ c) $7\frac{3}{4}$ d) $6\frac{3}{4}$

20) y equals the product of x and 3 is …….

a) $y = 3x$ b) $x = 3y$ c) $x = 3 + y$ d) $y = 3 + x$

21) Which of the following is an integer?

a) $\frac{5}{16}$ **b)** $-\frac{2}{4}$ **c)** $-\frac{15}{5}$ **d) -0.4**

22) Which of the following are relatively prime number?

a) 2, 6 b) 7, 9 c) 4, 8 d) 10,15

23) The number of integers between -5 and 2 is ……

a) 6 b) 5 c) 3 d) 7

24) The number of terms of expression $5x + 3y - 1$ is ……..

a) 5 b) 3 c) 1 d) 1

25) The mediam of 9, 4 8 1 and 3 is …….

a) 3 b) 4 c) 5 d) 8

26) {0} is the best subset of ……

a) Rational b) integer c) counting d) natural

27) The L.C.M of 5 and 7 is ……..

a) 1 b) 5 c) 7 d) 35

28) The range of 11, 7, 9, 13, 3 is …….

a) 9 b) 10 c) 8 d) 2

29) The range = the greatest value …….. the smallest value

a) - b) + c) × d) ÷

30) The mode of 7, 9, 8, 7, 6, 7,and 10 is ……..

a) 6 b) 7 c) 8 d) 9

31) Eight squared =……

a) 2×8 **b)** $8 + 2$ **c)** 8^2 **d)** $8 \div 2$

32) 1 is the best subset of ……

a) Rational number b) natural number

c) Integer d) counting number

33) (-4) ……0

a) $<$ b) $>$ c) $=$ d) $\geq$

34) Which display makes it easier to see the median?

a) Histogram b) box plot

c) dot plot d) bargraph

35) 8+24=8(……..+……)

a) 1,3 b) 8, 2 c) 3,2 d)1, 7

36) $-\frac{1}{4}$……$-\frac{1}{7}$

a) $<$ b) $>$ c) $=$

37) 6 times m added to 2 equal n in equation form is ……

a) 6m-2=n b) 6n+2=m c) 6m+2=n d) 6×2m=n

38) 2^3 =……..

a) 2×3 b) 3×3 c) $2\times 2 \times 2$ d) 2+3

39) Which of the following is an integer ……..

a) $\frac{16}{4}$ b) $-\frac{2}{4}$ c) $-\frac{15}{5}$ d) -0.4

40) Which of the following is an algebric expression?

a) $3^2 - 4$ b) $5x + 3$ c) $29 - 3^2$ d) 2(4+5)

41) In the equation 1860÷32=58 R4, the remainder is ……..

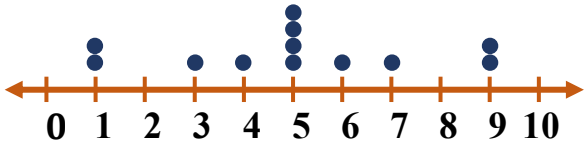
a) 1860 b) 32 c) 58 d) 4

42) …..is the middle value of the data et

a) mean b) mode c) median d) outlier

2) Complete:

1) $|4| \times |-4|$=……..
2) $9 \times 9 \times 9 \times 9 = 9$….
3) The smallest counting number is …….
4) 8-3×2÷(4-2)=……..
5) 8(5+40=40+……..
6) The distance between 5 and -5 on the number line is ……. Units
7) In equation $3410 \div 63 = 54$ R8 the divisor is…….
8) If the mean of 3, 7, 6, x is 5 then x =…….
9) In the equation L=4m-3, the independent variable is……..
10) The outlier value of the following data (203, 204, 205, 23, 206, 207, 202, 201) is ……..
11) L.C.M of 5 and 8 is……
12) The coefficient of 2+3a-5 is……
13) The mean of the following data is ……..



14) The L.C.M of 4 and 12 is ……
15) $7 \times 7 \times 7 = 7$…….
16) The constant in the expression 5x+2 is ……..
17) If m-2 =7, then m+1 =…….
18) $\frac{7}{9} - \frac{5}{9}$ =…….
19) $2 \times 3 + 5 \times 5$=……
20) the……=the greatest value-the smallest value
21) in the equation $7 + z = 17$, then te valueof z=…….
22) The coefficient in the algebric term (3xy) is ……
23) If 5=$|a|$ then a = ……or……
24) If (7x=35), then the value of x=……..
25) In the equation (y=x+4), then the dependent variable is ……..
26) The G.C.F of the two relatively prime numbers is ……
27) If the mean of the values (3, 4, 9, x, 8) is 6 then the value of x is ……
28) The outlier in the set of values (5, 18, 3, 4, 7, 6) is ……

3) Solve the following:

1) Use the order of mathematical operations to simplify:

$$40 + 5(3^2 - 7) + 10$$

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……
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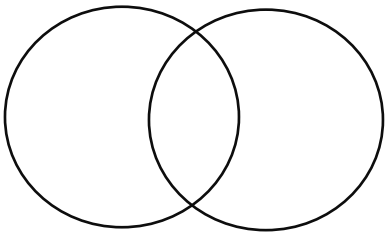
2) Order the given set of numbers from least to greatest:

2.1, $-3\frac{1}{4}$, $-1\frac{7}{8}$, $-2\frac{1}{2}$

The order:….., ….., ……, …….

3) Find the G.C.F. and L.C.M of 4 and 6 using venn diagram:

……
……
……
……
……



4) Complte the following table, then rpresent graphically the equation y=x+1

x	0	1	2
Y			
(x, y)	(0,)	(1,)	(2,)

……
……
……
……

5) Find G.C.F for 24 and 16

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……
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……

6) **A factory produces 48 pieces of cloth daily. How many days does it produce 1,152 pieces of cloth?**

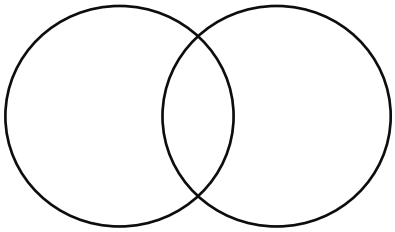
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7) **Use the ordr of operations to simplify:** $\mathbf{4 + 3^2 \times 2 \div (3 - 1)}$

……
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……

8) **Find the G.C.F and L.C.M of 10 and 30 using ven diagram**

……
……
……
……
……



9) **Complete the following table, then represent it graphically**
The equation: y=3x+1

x	0	1	2	3
Y				
(x, y)	(0,)	(1,)	(2,)	(3,)

……
……
……

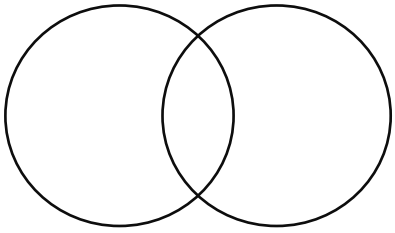
10) **Find the G.C.F and L.C.M using the venn diagram for 24 and 18**

24=……

16=……

G.C.F=…….

L.C.M=……..



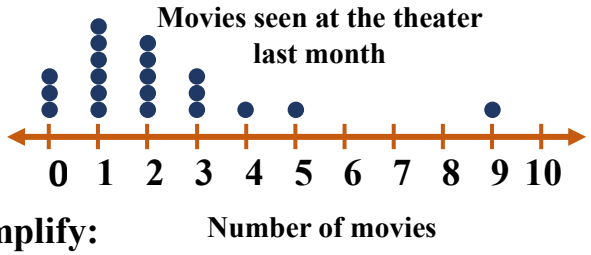
11) Ahmed has $5\frac{2}{3}$ and tamer has $15\frac{1}{2}$L.E, find out the total sum of what they have together.

……
……

12) Find the value of $3x2^3 \div 12$

……
……

13) from the opposite dot plotm answer the following questions:



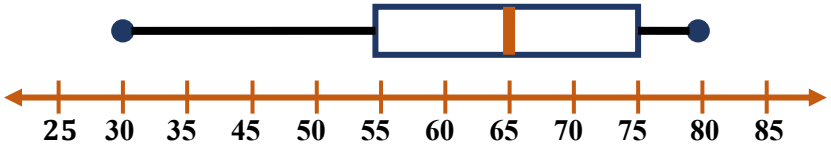
a) How many people saw 2 movies?
b) How many people surveyed?
c) How many people saw 3 movies or more?

14) use ordet of operations and exponent to simplify:

$$2 \times 4 + (3^2 - 20 \div 5)$$

……
……
……
……

15) use the opposite box plot to complete the following:



a) minimum value=……..
b) median=……
c) maximum value=……..
d) lower quartile=…….
e) upper quartile=…….

16) arrange in an ascending ordet:

5, -5,5 -5, 5.7, -1, $|-6|$

……., ……., ……., ……., ……., ……